

Article

Analysis of the Driving Stability of a Multi-Model Driven Electric-Tracked Intelligent Detection Device

Yitian Wang ^{1,*}, Huazhen Liu ², Linsen Song ^{1,*}, Longkai Liang ¹, Yiwen Zhang ¹, Zhenglei Yu ², Chunbai Liu ³, Yanzhong Hao ³, Meichao Xu ¹, Xinming Zhang ⁴ and Siying Li ⁵

- ¹ College of Mechanical and Electrical Engineering, Changchun University of Science and Technology, Changchun 130013, China; liangk2022@cust.edu.cn (L.L.); zhyw@cust.edu.cn (Y.Z.); xumeichao1991@yeah.net (M.X.)
- ² Key Laboratory of Bionic Engineering (Ministry of Education), Jilin University, Changchun 130022, China; liuhuazhenjlu@163.com (H.L.); zlyu@jlu.edu.cn (Z.Y.)
- ³ FAW Foundry Co., Ltd., Changchun 130013, China; liuchunbai@faw.com.cn (C.L.); haoyanzhong@faw.com.cn (Y.H.)
- ⁴ School of Mechatronic Engineering and Automation, Foshan University, Foshan 528225, China
- ⁵ School of International Economics and Trade, Jilin University of Finance and Economics, Changchun 130117, China; siyingli666@126.com
- * Correspondence: yitian18@cust.edu.cn (Y.W.); songlinsen@cust.edu.cn (L.S.)

Abstract: The electric-tracked intelligent detection device is extensively used in engineering exploration and construction. However, in unstructured and complex terrains, its stability is challenged by a high center of gravity and poor road conditions, making the enhancement of its stability analysis crucial. This study is based on tracked vehicle models, constructing kinematic and dynamic equations for the device, and developing virtual and physical prototypes. Combining theoretical analysis, virtual simulations, and physical testing, the study analyzes motion parameters under typical conditions through numerical calculations and simulations. It examines the device's dynamic characteristics during rapid turns in unstructured terrains. The research identifies variations in motion parameters, determines dynamic stability thresholds, and refines stability analysis methods applicable to the device, establishing evaluation criteria with multiple parameters and weight vectors. In conclusion, the developed models and prototypes help us understand the electric-tracked intelligent detection device's dynamic characteristics and instability thresholds, enhancing its stability and operational efficiency and providing a theoretical foundation for related engineering machinery development.

Keywords: electric-tracked intelligent detection device; kinematic and dynamic models; driving stability; virtual prototype simulation



Citation: Wang, Y.; Liu, H.; Song, L.; Liang, L.; Zhang, Y.; Yu, Z.; Liu, C.; Hao, Y.; Xu, M.; Zhang, X.; et al. Analysis of the Driving Stability of a Multi-Model Driven Electric-Tracked Intelligent Detection Device. *Actuators* **2024**, *13*, 527. <https://doi.org/10.3390/act13120527>

Academic Editor: Han Sol Kim

Received: 19 November 2024

Revised: 12 December 2024

Accepted: 16 December 2024

Published: 20 December 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The electric-tracked intelligent detection device, known for its exceptional off-road capabilities and maneuverability, is crucial in engineering exploration, mining, construction, and military survey [1]. However, as its application expands to more complex terrains like mountains and deserts, the high center of gravity due to various onboard instruments and detection equipment significantly reduces its stability. This is particularly problematic in rugged, unstructured environments where autonomous navigation can lead to frequent tipping incidents, thus decreasing the efficiency of resource exploration and transportation [2]. Therefore, researching ways to enhance the driving stability analysis of this device in unstructured environments is of significant theoretical and practical importance.

Over the years, as engineering vehicles have been widely utilized in various projects, researchers have conducted extensive and in-depth studies on methods for analyzing the stability of vehicle operations [3,4]. For instance, Lei T et al. analyzed factors affecting vehicle stability and their mechanisms using the eigenvalue method based on a

nonlinear dynamic model of articulated vehicles and their hydraulic steering systems [5]. Lai F et al. evaluated the stability of vehicle chassis control systems by analyzing the coupling between the vehicle's center of mass sideslip angle and yaw rate, using an 18-degree-of-freedom vehicle dynamics model to assess the impact of the front wheel steering angle, road adhesion coefficient, braking methods, and initial speed [6]. Zhou et al. employed lateral displacement, sideslip angle, lateral and longitudinal speeds, pitch angle, and fuel consumption to assess driving dynamic stability during vehicle following and lane changing [7]. Additionally, Zhu et al. used static bifurcation theory to determine vehicle stability regions influenced by engine braking torque, verifying stability through various initial longitudinal speeds and steering angles [8]. However, it is noteworthy that these studies primarily rely on numerical models for vehicle stability analysis, which may lead to incomplete and inadequate assessments.

With the significant advancements in computational and simulation capabilities, virtual prototyping simulation technology has shown promising applications in vehicle simulation, mechanical manufacturing, aerospace, and other fields. Liu et al. designed a trajectory tracking controller based on a four-channel adaptive disturbance rejection system using the Carsim platform, successfully analyzing the stability of four-wheel steering vehicles against crosswinds, skidding, and rollover [9]. Yu integrated virtual prototyping, nonlinear static tire models, and linear dynamic models for predictive control, effectively simulating vehicle dynamics under complex conditions to enhance the handling stability of four-wheel steering vehicles [10]. Nguyen et al., based on a high-fidelity vehicle dynamics control model, deeply analyzed yaw stability and path-tracking performance in adverse driving environments [11]. Zhou's team explored rollover mechanisms of high-center-of-gravity commercial vehicles using static equilibrium and phase plane analysis methods, achieving layered control of vehicle roll stability through adaptive model predictive control [12]. Kati et al. addressed the high-speed lateral stability of A-type double-trailer vehicles by introducing active steering technology for the front axle bogie, validating the controller's performance with a high-precision vehicle model [13].

In summary, despite the numerous factors contributing to vehicle instability, the resulting instability phenomena often exhibit common characteristics. For instance, key parameters such as yaw rate, roll acceleration, or vertical acceleration can lead to instability when they exceed predefined safety thresholds. By identifying these standard features, we can accurately predict the risk of vehicle instability through comparative analysis of driving stability metrics and corresponding boundary thresholds. This enables the timely activation of warning mechanisms and implementation of proactive safety control strategies in potentially hazardous traffic situations [14,15].

With ongoing advancements in vehicle testing technology, dynamic stability evaluation metrics continue to evolve, offering more precise descriptions of vehicle stability characteristics. Kenda J et al. analyzed vehicle cornering stability in sudden situations by considering maximum lateral acceleration, longitudinal acceleration, and braking force as evaluation metrics [16]. Dandiwala et al. optimized vehicle stability control across a broad speed range, targeting minimal perceived lateral acceleration and tilt torque as critical indicators [17]. Steinga developed a multibody dynamics model for specialized trailer applications to simulate dynamic performance during high-speed turning and lane changes, using Static Roll Threshold (SRT) and Load Transfer Ratio (LTR) to quantify roll stability [18]. Fu et al. explored the effects of turn rate, center of mass height, and road adhesion on the rollover stability of heavy automated guided vehicles, using the lateral deflection angle of the center of mass as a critical evaluation metric [19]. Wang et al. established a rollover assessment system based on nonlinear suspension theory and planar vehicle dynamics, using LTR as the core evaluation metric for rollover risk, successfully preventing vehicle rollover [20]. Additionally, Sukumaran et al. used a three-degree-of-freedom dynamics model to select LTR as the benchmark for assessing rollover risk in heavy commercial road vehicles, enhancing safety by integrating anti-rollover and anti-lock braking systems (ABS) [21]. However, electric-tracked intelligent detection devices

need help maintaining safety and stability during rapid travel due to their high center of gravity and operation on unstructured, harsh, and inclined terrains. Current stability evaluation methods still have limitations, making it difficult for these devices to achieve stable behavior during autonomous operation and emergency obstacle avoidance.

Based on the studies above, the specific research content is as follows: Firstly, we focused on tracked vehicles' kinematics and dynamics models to develop both virtual and physical prototype models. Next, we conducted simulation validation of the virtual prototype model for the electric-tracked intelligent detection device. Finally, through theoretical modeling and multi-model-driven virtual prototype simulation, we thoroughly analyzed the device's dynamic characteristics during rapid maneuvers under unstructured adverse conditions, providing a solid theoretical foundation for the stability control of tracked engineering vehicles.

2. Kinematic Modeling of Electric-Tracked Intelligent Detection Device

2.1. Kinematic Analysis

We utilize differential steering to control the turning of the electric-tracked intelligent detection device. Assuming the entire track system is a rigid body with no flexible deformation and both tracks have the same lateral slip, the device's movement in the ground coordinate system is illustrated in Figure 1.

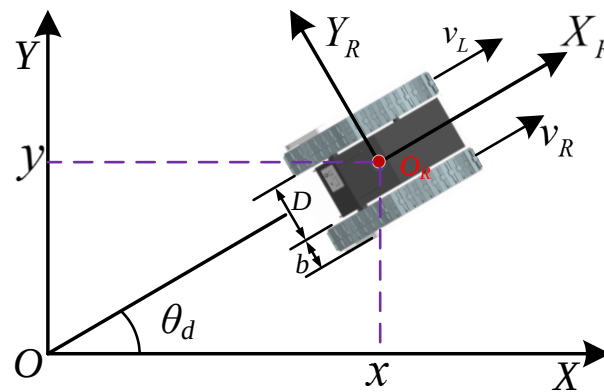


Figure 1. Schematic diagram of the motion analysis of an electric-tracked intelligent detection device.

Establish the ground coordinate system XOY and the device coordinate system $X_R O_R Y_R$, where O_R represents the centroid of the detection device. The $O_R X_R$ direction is the forward direction of the device, while $O_R Y_R$ is lateral to the tracks and perpendicular to $O_R X_R$. The linear velocities of the left and right tracks are v_R and v_L , respectively, aligned with $O_R X_R$. Angle θ_d is the orientation between the ground and device coordinate systems (positive for counterclockwise rotation and negative for clockwise). D represents the body width, and b is the track width [22].

The orientation equation of the detection device at time t can be expressed as follows:

$$\dot{W} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta}_d \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \cos \theta_d & \frac{1}{2} \cos \theta_d \\ \frac{1}{2} \sin \theta_d & \frac{1}{2} \sin \theta_d \\ -\frac{1}{D+2b} & \frac{1}{D+2b} \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix} \quad (1)$$

Based on the characteristics of the track structure, the linear velocity v_o and angular velocity ω_o of the track's centroid O_R can be derived from the linear velocities v_R and v_L of the left and right tracks, as shown in Equation (2):

$$\begin{bmatrix} v_o \\ \omega_o \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{D+2b} & \frac{1}{D+2b} \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix} \quad (2)$$

Let the turning radius of the track be R . Based on the formula $v = R \cdot \omega$ and Equation (2), the specific calculation formula for the turning radius can be derived as follows:

$$R = \frac{v_O}{\omega_O} = \frac{D + 2b}{2} \cdot \frac{v_L + v_R}{v_L - v_R} \quad (3)$$

By substituting Equation (2) into Equation (3), the position and orientation equation of the detection device can be updated, where H is the velocity state matrix of the detection device in the ground reference coordinate system:

$$\dot{W} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta}_d \end{bmatrix} = \begin{bmatrix} \cos \theta_d & 0 \\ \sin \theta_d & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_O \\ \omega_O \end{bmatrix} = H \begin{bmatrix} v_O \\ \omega_O \end{bmatrix} \quad (4)$$

2.2. Kinematic Model for Straight-Line Motion

When $v_R = v_L$, the linear velocities of the left and right tracks' centroids are equal, and the detection device maintains straight-line motion. According to the pose state equation of the detection device, the rate of change of the orientation angle at time t is as follows:

$$\dot{\theta}_d = 0 \quad (5)$$

At this time, the pose equation of the detection device is as follows:

$$\begin{cases} x(t) = \frac{1}{2} \int_0^t (v_L + v_R) \cos \theta_0 dt = v_L t \cos \theta_0 \\ y(t) = \frac{1}{2} \int_0^t (v_L + v_R) \sin \theta_0 dt = v_L t \sin \theta_0 \\ \theta_d(t) = \theta_0 \end{cases} \quad (6)$$

2.3. In-Place Rotation Kinematic Model

When $v_R = -v_L$, the linear velocities of the centroids of the left and right tracks are equal in magnitude but opposite in direction, causing the detection device to perform an in-place rotation. According to the pose state equation of the detection device, the rate of change of the orientation angle at time t is as follows:

$$\dot{\theta}_d = \frac{2v_L}{D + 2b} \quad (7)$$

From the above equation, the position and orientation equations of the detection device can be derived as follows:

$$\begin{cases} x(t) = \frac{1}{2} \int_0^t (v_L + v_R) \cos \theta_0 dt = 0 \\ y(t) = \frac{1}{2} \int_0^t (v_L + v_R) \sin \theta_0 dt = 0 \end{cases} \quad (8)$$

Thus, the orientation angle θ_d of the detection device at the current time is as follows:

$$\theta_d(t) = \frac{1}{D + 2b} \int_0^t (-v_L + v_R) dt = \frac{-2v_L}{D + 2b} \cdot t \quad (9)$$

2.4. Differential Steering Kinematic Model

When $v_R \neq v_L$ and $v_L \neq -v_R$, assuming a constant forward speed of the detection device during turning, the velocity difference between the track centers $\Delta v = v_L - v_R$ is steady, indicating the device is performing differential steering.

The rate of change of the orientation angle of the detection device at time t is as follows:

$$\dot{\theta}_d = \frac{\Delta v}{D + 2b} \quad (10)$$

According to Equation (10), the motion state equation of the detection device at the current time is as follows:

$$\begin{cases} x(t) = \frac{1}{2} \int_0^t (v_L + v_R) \cos \theta_d dt = \frac{2v_R + \Delta v}{2} \times \frac{D + 2b}{\Delta v} \times \sin\left(\frac{\Delta v}{D + 2b} \times t\right) \\ y(t) = \frac{1}{2} \int_0^t (v_L + v_R) \sin \theta_d dt = \frac{2v_R + \Delta v}{2} \times \frac{D + 2b}{\Delta v} \times \left[1 - \cos\left(\frac{\Delta v}{D + 2b} \times t\right)\right] \\ \theta_d(t) = \frac{\Delta v}{D + 2b} \times t \end{cases} \quad (11)$$

Based on $\sin^2(x) + \cos^2(x) = 1$, Equation (11) can be transformed into the standard form of a circle's equation:

$$x^2(t) + \left[y(t) - \frac{(2v_R + \Delta v)(D + 2b)}{2\Delta v}\right]^2 = \left[\frac{(2v_R + \Delta v)(D + 2b)}{2\Delta v}\right]^2 \quad (12)$$

According to Equation (12), when the movement directions of the left and right track subsystems of the detection device are the same, and the speed difference between the tracks is constant, the coordinates of the turning center are as follows:

$$\left(0, \frac{(2v_R + \Delta v)(D + 2b)}{2\Delta v}\right) \quad (13)$$

The turning radius of the electric-tracked intelligent detection device can be expressed as follows:

$$R = \frac{(2v_R + \Delta v)(D + 2b)}{2\Delta v} = \frac{(v_R + v_L)(D + 2b)}{2(v_R - v_L)} \quad (14)$$

3. Dynamic Model of an Electric-Tracked Intelligent Detection Device

During autonomous operation, the electric-tracked intelligent detection device must overcome multiple resistive forces, including travel resistance F_f , inertia resistance F_i , climbing resistance, and wind resistance. Based on idealized assumptions, we select straight-line and turning conditions to establish the corresponding dynamic model of the tracked mechanism.

3.1. Force Analysis of Dynamic Model

3.1.1. Travel Resistance F_f

By analyzing the load, the travel resistance caused by friction and deformation of the track during operation can be calculated as follows:

$$F_f = f_g \cdot m \cdot g \quad (15)$$

f_g represents the rolling resistance coefficient in this formula, which varies on different surfaces. Given that electric-tracked intelligent detection devices are commonly used in muddy outdoor environments, we set the rolling resistance coefficient f_g to 0.1.

3.1.2. Inertial Resistance F_i

The inertial resistance experienced by the detection device at the moment of startup can be calculated using the following formula:

$$F_i = m \cdot a = m \cdot \frac{v}{t} \quad (16)$$

In this formula, a represents the acceleration of the device's center of mass, v is its linear velocity, and t denotes the time required for acceleration.

3.1.3. Gradient Resistance F_s

As shown in Figure 2, the gradient resistance encountered by the detection device while traveling uphill can be calculated based on the slope angle.

$$F_s = G_i \sin \alpha \quad (17)$$

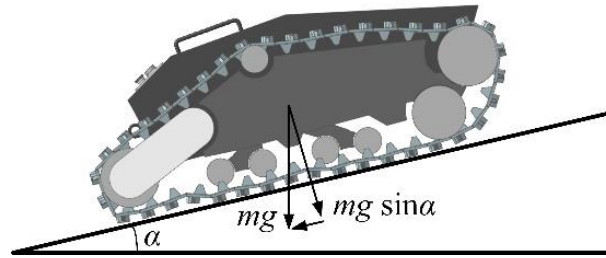


Figure 2. Force analysis of the slope-climbing process for the electrically driven tracked intelligent detection device.

When the detection device travels on roads with a slope angle of less than 20%, the gradient resistance can be calculated using the following formula:

$$F_* = F_\theta \times \tan \dot{\alpha} = G_i \times \frac{P}{100} \quad (18)$$

P represents the percentage gradient, and the formula's calculation error does not exceed 2%. Although climbing conditions slightly affect frontal resistance, this impact is generally negligible.

3.1.4. Aerodynamic Resistance F_w

The aerodynamic resistance experienced by the detection device can be calculated using the following formula:

$$F_w = \frac{\rho_l}{2} \cdot v_{rel}^2 \cdot C_w \cdot A_w \quad (19)$$

In this formula, ρ_l denotes air density; v_{rel} represents the relative velocity between the device and air; C_w is the drag coefficient; and A_w is the frontal area of the device. Since aerodynamic resistance is relatively minor compared to other forces, it can often be neglected in calculations.

In summary, the driving force required to initiate the movement of the detection device on flat and inclined surfaces can be expressed as follows:

$$F_Q = F_s + (F_f + F_{\max} + F_w/2) \cos \alpha + \frac{m}{2} \dot{v}_c \quad (20)$$

$$F_Q = F_i + F_{\max} + F_f + \frac{m}{2} \dot{v}_c \quad (21)$$

By conducting a force analysis of the detection device under different operating conditions, we can derive the dynamic equations for a single track during straight movement on flat terrain and uphill travel, as shown in Equations (22) and (23). Formula (22) describes the situation of the center of mass linear velocity and center of mass angular velocity of a single-sided track when the detection device is traveling in a straight line on a flat road. Considering that the detection device is in a straight-line state at this time, its center of mass angular velocity is 0. Formula (23) corresponds to the center of mass linear velocity and center of mass angular velocity of the single-sided track when the detection device

travels in a straight uphill line. Due to the existence of slope, the uphill resistance F_s is introduced in Formula (23).

$$\begin{cases} \dot{v}_c = \frac{2[F_Q - (F_i + F_f + F_{\max} + F_w/2)]}{m} \\ \dot{\omega}_c = 0 \end{cases} \quad (22)$$

$$\begin{cases} \dot{v}_c = \frac{2\{F_Q - [F_s + (F_f + F_{\max} + F_w/2) \cos \alpha]\}}{m} \\ \dot{\omega}_c = 0 \end{cases} \quad (23)$$

3.2. Differential Steering Dynamics Model

R represents the turning radius of the electrically driven tracked intelligent detection device around point O . Additionally, the device's driving and braking forces are closely related to the turning radius. L denotes the overall length of the track chassis, B is the track gauge of the detection device, and V_1 and V_2 represent the linear velocities of the centroids of the tracks on each side, as illustrated in Figure 3 [23].

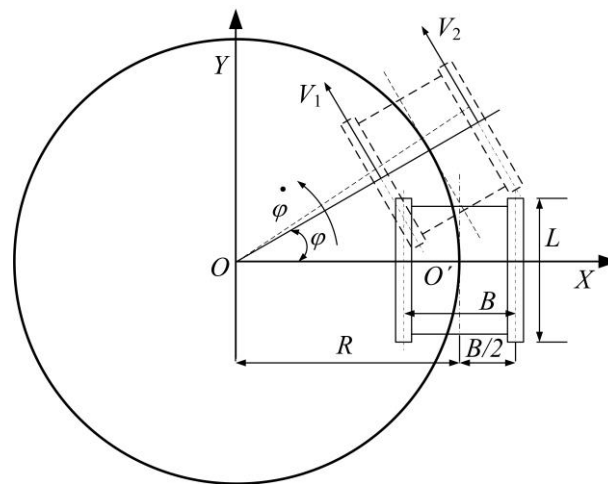


Figure 3. Schematic analysis of the steering mechanism for an electrically driven tracked intelligent detection device on level ground.

Considering the relatively smooth steering process of the electrically driven tracked intelligent detection device, it can be assumed that the tracks are in a state of force equilibrium during maneuvering. In this state, the tracks are subjected to both driving and frictional forces from the ground. Figure 4 presents the force analysis of the track system while steering on a flat surface.

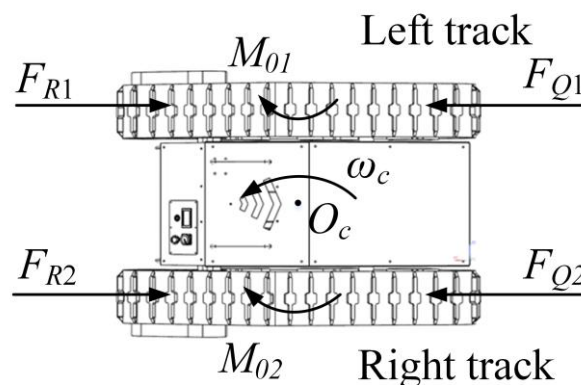


Figure 4. Force analysis diagram of the track system during steering on a flat surface.

When the detection device requires adjusting its movement posture during the startup phase and straight-line motion, the tracking subsystem undergoes a dynamic stabilization

process. Based on the force equilibrium analysis diagram for track steering shown in Figure 5, the dynamic analysis equation for track steering can be derived, as shown in Equation (24):

$$\begin{cases} F_{Q2} + F_{Q1} - F_{R1} - F_{R2} = m\dot{v}_c \\ (F_{Q2} - F_{Q1})B/2 - M - (F_{R2} - F_{R1})B/2 = J_O\dot{\omega}_c \end{cases} \quad (24)$$

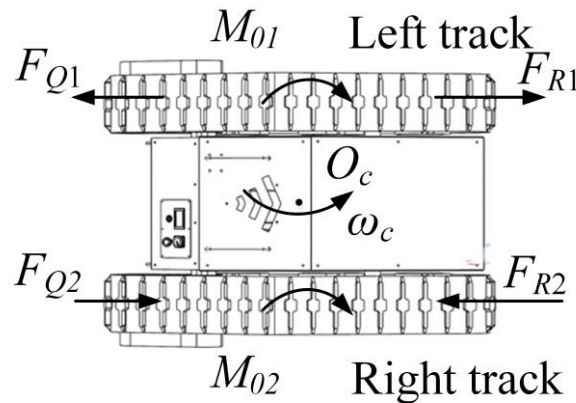


Figure 5. Schematic diagram of track force analysis during in-place steering conditions.

In this equation, the formula for calculating rolling friction resistance is $F_{R1} = F_{R2} = mgf_g/2$, where J_o is the moment of inertia of the detection device around its center of mass, m is the total mass of the device, and M is the rotational resistance torque of the detection device. Assuming no slippage or skidding of the tracks on either side, the rotational resistance torque can be calculated using Equation (25). R is the theoretical turning radius of the track, and R_k is the actual turning radius of the track (caused by the clearance and elasticity of the track). During the actual turning process, due to the occurrence of slip and slippage at the track contact area, as well as the influence of factors such as track clearance and elasticity, the actual turning radius may differ from the theoretical turning radius (assuming that the slip and slippage at the track contact area are zero).

$$\begin{cases} M = \mu mg l / 4 \\ \mu = \frac{\mu_{\max}}{(1 + \frac{2R}{B})^{n_k}} \left(1 - \frac{R}{R_k} \right) \end{cases} \quad (25)$$

Therefore, the formula for calculating the driving force required for the steering of the detection device can be determined as follows:

$$\begin{cases} F_{Q2} = F_{R2} + M/B + J_o\dot{\omega}_c/B + m\dot{v}_c/2 \\ F_{Q1} = F_{R1} - M/B - J_o\dot{\omega}_c/B + m\dot{v}_c/2 \end{cases} \quad (26)$$

Thus, based on the turning principle of the tracked walking mechanism, the dynamic equation for the steering motion of the detection device can be derived as follows:

$$\begin{cases} \dot{v}_c = \frac{F_{Q1} + F_{Q2} - (F_{R1} + F_{R2})}{m} \\ \dot{\omega}_c = \frac{B[-F_{Q1} + F_{Q2} - (-F_{R1} + F_{R2})] - 2M}{2J_O} \end{cases} \quad (27)$$

3.3. Dynamic Model for In-Place Steering Motion

When the detection device performs in-place steering around its geometric center, the turning radius is $R = 0$. At this point, the primary external forces on the tracks are the frictional resistance from the ground and the driving force applied by the system, as illustrated in Figure 5.

Based on the force analysis schematic for in-place track steering, the dynamic analysis equations for the track subsystems during the transition from startup to steady state can be derived, as shown in Equation (28):

$$\begin{cases} F_{Q2} - F_{Q1} + F_{R1} - F_{R2} = 0 \\ (F_{Q2} + F_{Q1})B/2 - M - (F_{R2} + F_{R1})B/2 = J_O\dot{\omega}_c \end{cases} \quad (28)$$

Equation (29) represents the driving force equation when the detection device is undergoing in-place steering conditions:

$$\begin{cases} F_{Q2} - F_{Q1} + F_{R1} - F_{R2} = 0 \\ (F_{Q2} + F_{Q1})B/2 - M - (F_{R2} + F_{R1})B/2 = J_O\dot{\omega}_c \end{cases} \quad (29)$$

The dynamics equation for in-place steering of the electrically driven tracked intelligent detection device is presented in Equation (30):

$$\begin{cases} \dot{v}_c = 0 \\ \dot{\omega}_c = \frac{(F_{Q2} + F_{Q1})B/2 - M - (F_{R2} + F_{R1})B/2}{J_O} \end{cases} \quad (30)$$

4. Experimental Validation of the Kinematic Model

4.1. Model Development

As shown in Figure 6, based on the design dimensions of the experimentally tracked vehicle under challenging ground conditions, both a virtual prototype and a physical prototype were developed. The detection device's control and data acquisition system consist of a drive controller, a central controller, a vehicle speed sensor, an inertial measurement unit, and an accelerometer. The mobility system includes two sets of longitudinally arranged tracks, each powered by induction motors, with a maximum linear velocity of 2 m/s. The virtual prototype also incorporates rigid body models of components such as drive wheels and track plates. The main technical parameters of the detection device are listed in Table 1.

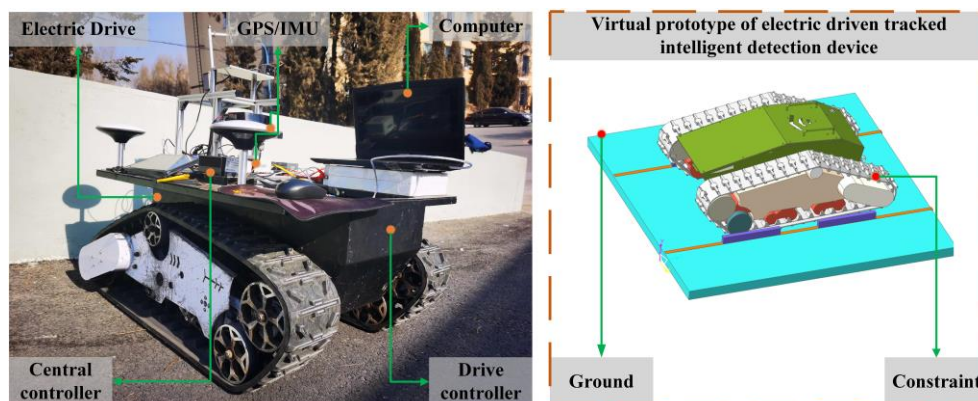


Figure 6. Electronically tracked intelligent detection device: physical and virtual prototypes.

Table 1. Main technical parameters of the electrically driven tracked intelligent detection device.

Parameter	Unit	Numerical Value
The total mass of the detection device	Kg	200
Track shoe width	m	0.18
Width of the detection device	m	0.38
Total length of track chassis	m	1.2
Track grounding length	m	0.85
Track Height	m	0.3

Table 1. Cont.

Parameter	Unit	Numerical Value
Track tension coefficient	-	0.35
Reduction ratio	-	100
Ground pressure ratio	Mpa	0.045
Maximum steering resistance Coefficient	-	0.6
Drive wheel radius	m	0.16
Driven wheel radius	m	0.15
Bearing wheel radius	m	0.14

4.2. Experimental Design

We conducted virtual prototype experiments, physical prototype experiments, and numerical simulation verification of kinematic models for four typical operating conditions: straight driving on flat roads, turning driving on flat roads, straight driving on sloping roads, and turning driving on sloping roads [24]. A virtual prototype simulation model was established through model design, setting parameters and constraints, and setting simulation step sizes; the data collection of the physical prototype experiment is completed through the inertial measurement unit, acceleration sensor, and vehicle speed collector, and the motion control of the detection device is completed through the central controller, and an improved Kalman algorithm is adopted to improve measurement accuracy [25]. We evaluated the accuracy of the kinematic model of the electric driven tracked intelligent detection device by comparing the deviations in centroid velocity, acceleration, heading angle, and heading angular velocity between actual and ideal environments. The experimental process is shown in Figure 7.

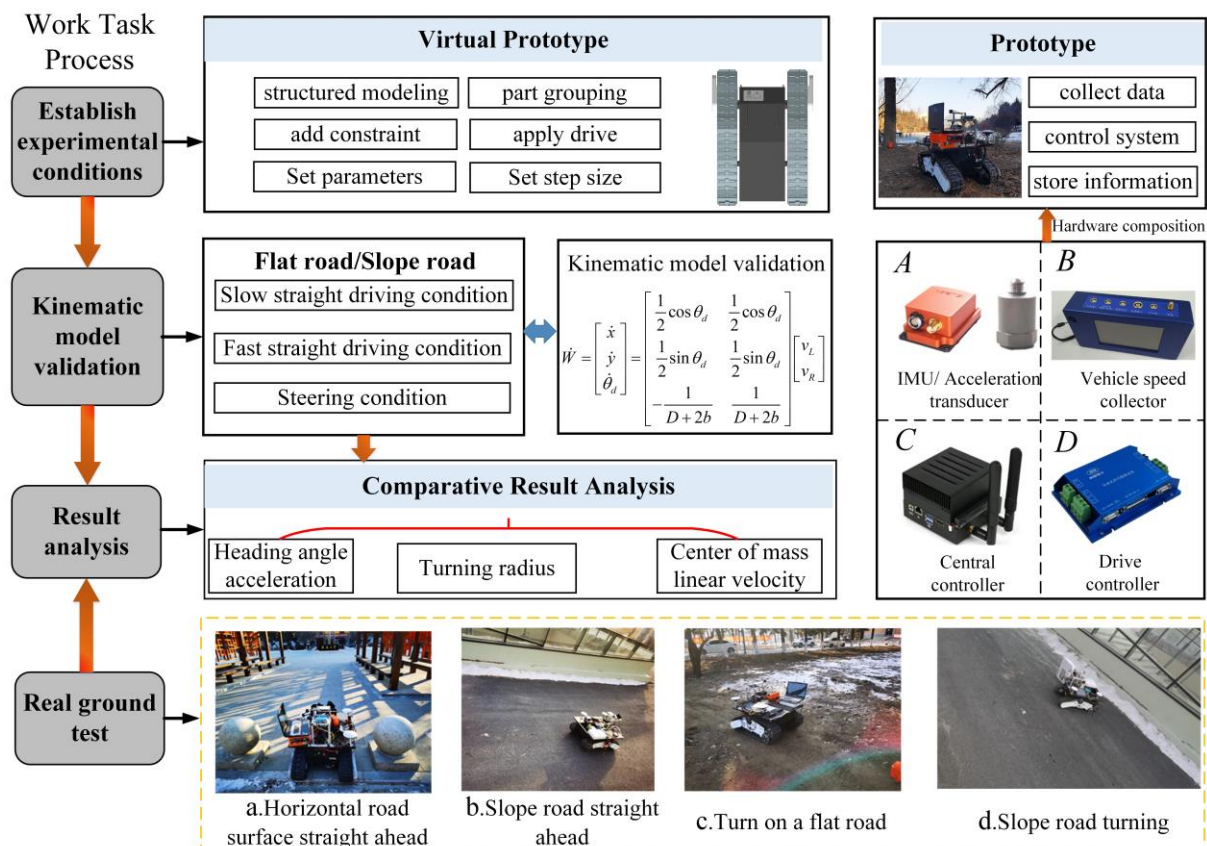


Figure 7. Task allocation diagram for experimental conditions of the electrically driven tracked intelligent detection device.

4.3. Results and Discussion

Given that this detection device mainly operates in unstructured outdoor environments such as mountains and sandy areas, we can change the properties of the road surface by adjusting ground parameters to more accurately simulate the motion and performance of the detection device under complex road conditions. The specific settings of ground parameters, such as ground stiffness, cohesion, shear angle, and shear deformation, are detailed in Table 2.

Table 2. Table of ground contact parameters for field conditions.

Parameter	Unit	Numerical Value
Ground stiffness	k_c	4.7613×10^{-4}
Cohesiveness	c	1.04×10^{-3}
Angle of shear	-	28
Shear deformation modulus	k	25

4.3.1. Straight Motion on Horizontal Surface Conditions

The detection device moves forward on a horizontal surface at stable speeds of 0.5 m/s and 1.5 m/s. Validation results are derived by comparing numerical calculations from the kinematic model, simulation outcomes from the virtual prototype, and results from the physical prototype. The comparison includes numerical values for the center of mass velocity, center of mass acceleration, and heading angle.

Figure 8a–d illustrate the comparison results of the center of mass velocity, center of mass acceleration, and heading angle for the electrically driven tracked intelligent detection device. Figure 8 shows that the kinematic model solutions and virtual prototype simulation results closely align with the physical prototype experiment results. The kinematic model and simulation results are smoother, while the physical prototype results show significant fluctuations. This is partly due to the uneven surface of the actual test environment compared to the ideal soft surface in simulations. Additionally, the virtual prototype assumes ideal constraints between components, while the physical prototype has assembly errors, causing relative motion and vibration during operation. This results in inertial measurement data fluctuations, although the overall data trend remains consistent.

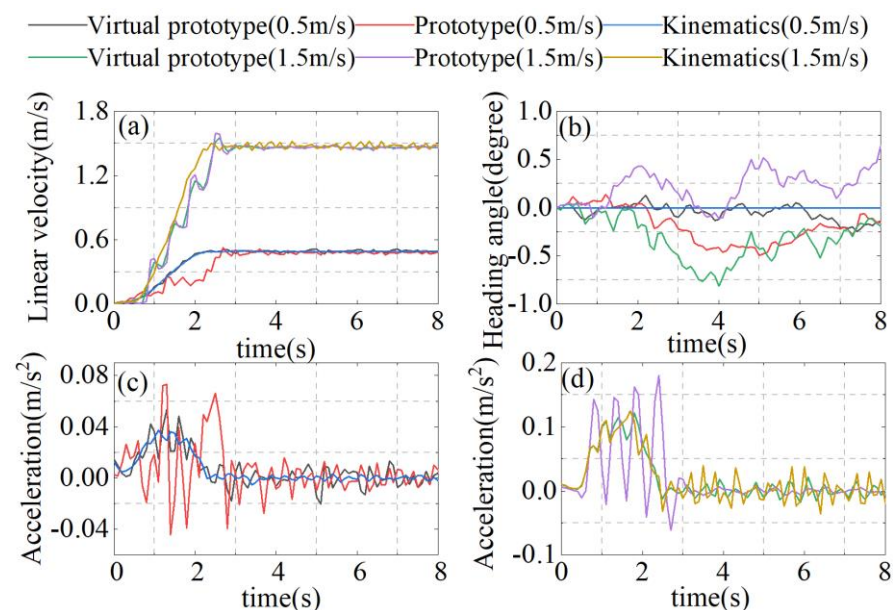


Figure 8. Comparison of experimental results on a horizontal surface. (a) Comparison of Linear velocity. (b) Comparison of Heading angles. (c) Acceleration comparison. (d) Acceleration comparison.

In comparing straight-line movement on a flat road, the detection device travels at stable speeds of 0.5 m/s and 1.5 m/s. Figure 8a,c,d show that within 0–3 s, the device is in the acceleration phase, displaying significant speed errors. Between 3 and 8 s, the device reaches steady-state motion, where the consistency of the center of mass velocity among the virtual prototype, physical prototype, and theoretical calculations is high, with acceleration changes remaining stable. The average error in the center of mass velocity between the virtual and physical prototypes is 6.5%, meeting current engineering requirements. Figure 8b indicates that the heading angle error across the three methods remains within $\pm 0.6^\circ$, meaning the device maintains a straight path without deviating from the preset course. Acceleration comparisons are shown in Figure 8c,d. Table 3 presents the speed error between the physical prototype experiment and theoretical calculations during straight-line motion on a horizontal surface.

Table 3. Table of speed error analysis for straight-line motion on a horizontal surface.

Process	Average Relative Error	Mean Squared Error	Root Mean Square Error
Non-steady state driving process (0.5 m/s)	16.17%	0.48	0.69
Steady-state driving process (0.5 m/s)	3.03%	0.017	0.13
Non-steady state driving process (1.5 m/s)	12.69%	0.011	0.1
Steady-state driving process (1.5 m/s)	1.58%	0.0009	0.03

4.3.2. Conditions for Straight-Line Motion on an Inclined Surface

This section also analyzes the detection device's results under straight-line motion on an inclined surface. The device moves at stable speeds of 0.8 m/s and 1.6 m/s. Verification results are obtained by comparing the numerical calculations from the kinematic model, simulation results from the virtual prototype, and physical prototype results. These include values for the center of mass linear velocity, the center of mass acceleration, and the heading angle.

Due to limitations, this section conducts straight-line motion experiments on a 5° incline, designing scenarios for the direct start of the detection device. Figure 9a–d show good agreement between experimental and theoretical results. In Figure 9a, comparing the center of mass linear velocity curves reveals significant differences during acceleration between the physical prototype and virtual simulation/theoretical results. This is because the virtual simulation cannot fully replicate the startup acceleration of the physical prototype. The comparison of acceleration values of the detection device is shown in Figure 9c,d with the underlying mechanisms discussed earlier.

As shown by the yaw angle curve of the detection device in Figure 9b, a larger yaw angle occurs during the uphill process compared to previous scenarios. This is due to the need to overcome gravitational and frictional resistance to gain sufficient acceleration on the slope. The incline and surface roughness influence the friction, which may cause the device to experience lateral slipping during startup. Faster speeds increase the likelihood of yaw. Experimental results and empirical knowledge indicate that more significant inclines exacerbate yaw. However, at a speed of 1.6 m/s, both virtual simulations and physical experiments show the device's maximum absolute yaw angle is 1.3° , meeting current engineering requirements.

Table 4 illustrates the speed discrepancies between physical prototype tests and theoretical calculations for straight-line motion on inclined roadways. Compared to flat road conditions, the speed error during startup is more significant on inclined roads, primarily due to the detection device's lateral slipping when starting directly on the slope.

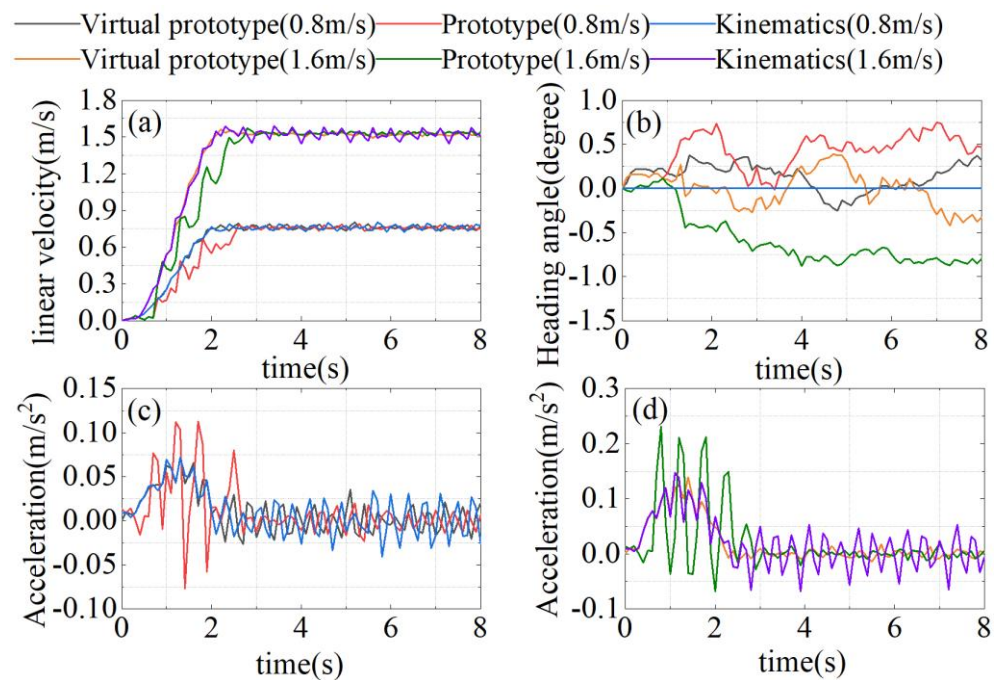


Figure 9. Comparison of experimental results on inclined surface. (a) Comparison of Linear velocity. (b) Comparison of Heading angles. (c) Acceleration comparison. (d) Acceleration comparison.

Table 4. Table of speed error analysis for straight-line motion on an inclined surface.

Process	Average Relative Error	Mean Squared Error	Root Mean Square Error
Non-steady state driving process (0.8 m/s)	13.32%	0.002	0.047
Steady-state driving process (0.8 m/s)	3.94%	0.0006	0.024
Non-steady state driving process (1.6 m/s)	12.69%	0.015	0.12
Steady-state driving process (1.6 m/s)	1.58%	0.008	0.089

4.3.3. Turning Conditions on Flat Roads

To validate the turning strategy of the detection device in the kinematic model, we conducted a detailed analysis of its turning performance on flat roads. For clarity, both virtual prototype simulations and physical prototype test results focused on the turning process. In a straight-line state, the relationship between the inner track centroid velocity v_1 and the outer track centroid velocity v_2 is $v_1 = v_2 = v_s$. During turning, the inner track velocity v_1 decreases while the outer track velocity v_2 increases, creating a velocity difference $\Delta v = v_2 - v_1$. The centroid velocities of both tracks are presented in Table 5.

Table 5. Centroid velocities of detection device tracks (m/s).

Process	Inner Track Line Velocity v_1	Outer Track Line Velocity v_2	Speed Difference Δv
1	1.3	1.1	0.2
2	1.4	1.0	0.4
3	1.5	0.9	0.6
4	1.6	0.8	0.8
5	1.7	0.7	1.0
6	1.8	0.6	1.2

As shown in Figure 10, both the theoretical turning radius and the actual turning radius of the physical prototype decrease with the increase in speed difference between the tracks. This occurs because a more significant speed difference (Δv) requires more turning torque to complete the maneuver, reducing the turning radius. The theoretical turning

radius is calculated using a simplified model. However, due to the complexity of the electric-driven tracked intelligent detection device's structure and motion, discrepancies exist between the actual and theoretical values, with an average relative error of 0.07 m, which meets engineering requirements. This result can be used to optimize the device's steering strategy further, enhancing its turning performance and stability.

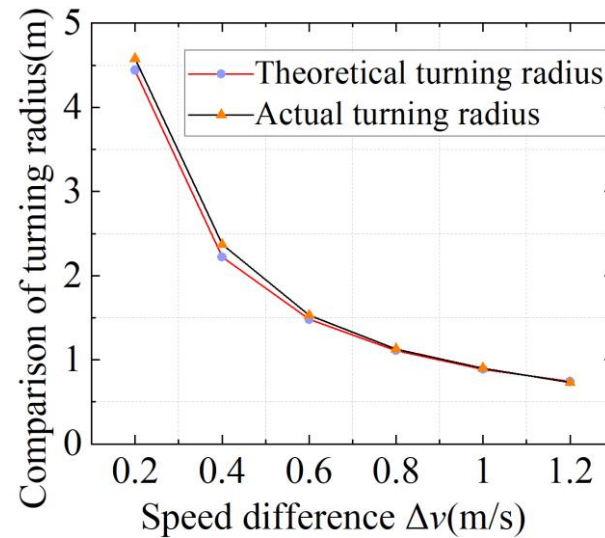


Figure 10. Comparison of theoretical and actual turning radius on flat terrain for the detection device.

The experimental results are illustrated in Figure 11, which compares the yaw and relative error rates of the detection device's center of mass during steady-state turning under varying speed differentials of 0.2, 0.4, 0.6, 0.8, 1.0, and 1.2 m/s between the two tracks. Figure 11a shows how the angular velocity of the detection device changes with the speed differential, specifically at a center of mass linear velocity of 1.2 m/s. It is evident from this figure that the results from all three approaches exhibit a similar trend; as the speed differential increases, the yaw rate also increases.

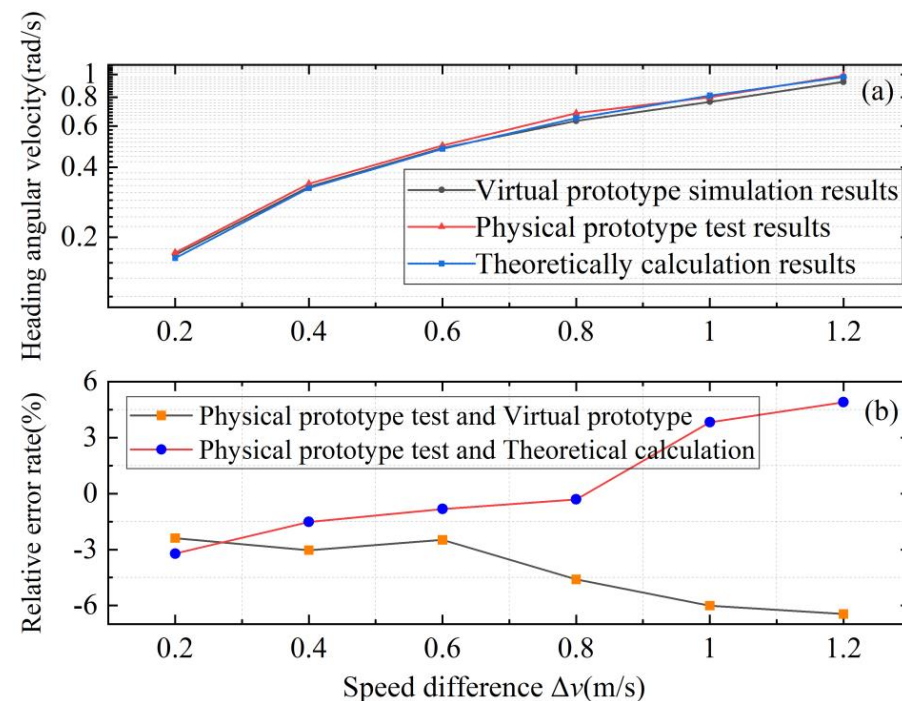


Figure 11. Comparison of experimental results for flat terrain steering conditions. (a) Comparison of heading angular velocity. (b) Relative error rate.

Figure 11b presents the comparative error between the virtual prototype simulation, physical prototype testing, and theoretical model calculations. The relative error among the three methods increases with the speed differential, with a maximum relative error of 6.4%. This error level is within a reasonable range, meeting current engineering requirements. These findings validate the high accuracy and stability of the kinematic model under steady-state turning conditions on flat terrain.

4.3.4. Turning Conditions on Slope Roads

This study analyzed the turning process of the electrically driven tracked intelligent detection device on slopes of 5° and 10° to validate its slope steering strategy. Theoretical and actual turning radii were compared, and the track centerline velocities are shown in Table 6.

Table 6. Centerline velocities of detection device tracks (m/s).

Slope	Process	Inner Track Line Velocity v_1	Outer Track Line Velocity v_2	Speed Difference Δv
5°	1	1.1	0.9	0.2
	2	1.2	0.8	0.4
	3	1.3	0.7	0.6
	4	1.4	0.6	0.8
	5	1.5	0.5	1
	6	1.6	0.4	1.2
10°	7	1.4	1.2	0.2
	8	1.5	1.1	0.4
	9	1.6	1.0	0.6
	10	1.7	0.9	0.8
	11	1.8	0.8	1
	12	1.9	0.7	1.2

Figure 12 shows that on a 10° slope, the actual turning radius is larger than the theoretical value. This is due to the device needing to overcome gravitational effects, requiring greater torque for turning. The average relative errors between theoretical and actual turning radii on 5° and 10° slopes are 0.06 m and 0.09 m, respectively, which fall within acceptable engineering limits.

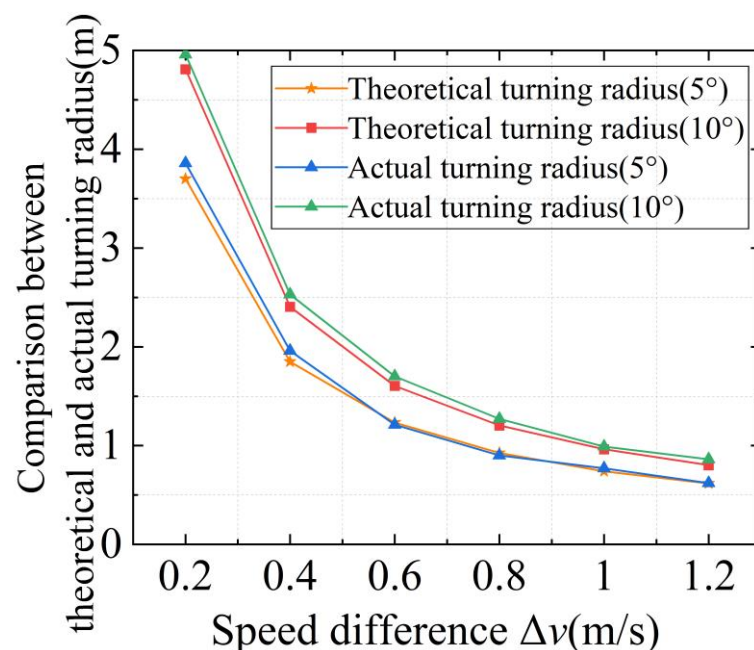


Figure 12. Comparison of theoretical and actual turning radii of the detection device on slopes.

The experimental results indicate that under varying slope conditions, the heading angular velocity of the detection device's center of mass, as obtained from virtual prototype simulation, physical prototype tests, and theoretical models, significantly changes with the increase in track speed difference (as shown in Figures 13 and 14). The relative error rate is minimized at a speed difference of 0.2 m/s and maximized at 1.2 m/s. These findings offer valuable insights into the steady-state turning performance of electrically driven tracked intelligent detection devices.

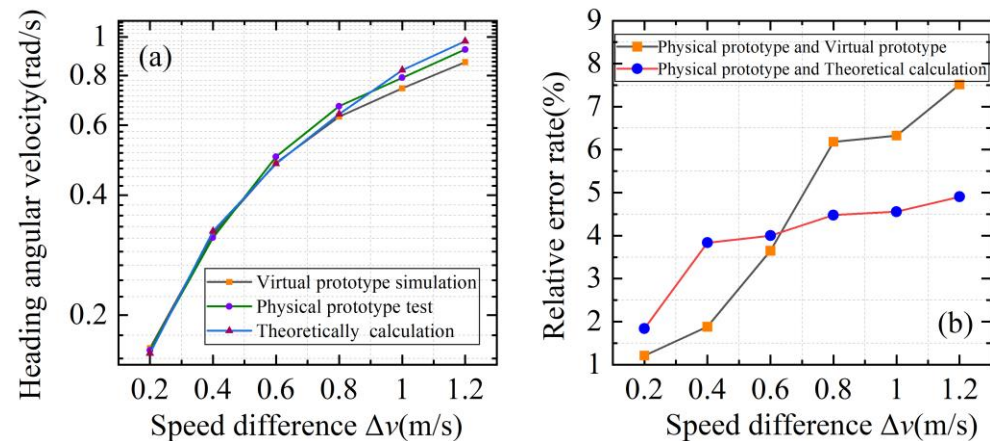


Figure 13. Comparison of experimental results for turning on a 5° slope. (a) Comparison of heading angular velocity. (b) Relative error rate.

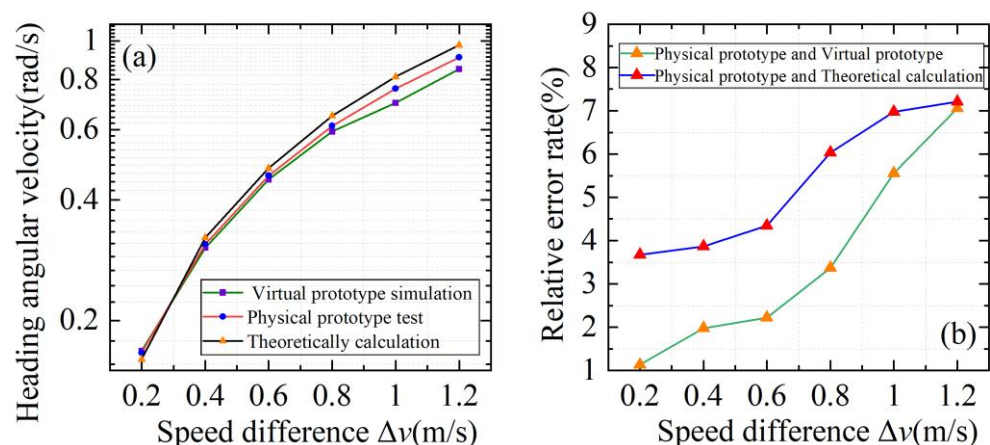


Figure 14. Comparison of experimental results for turning on a 10° slope. (a) Comparison of heading angular velocity. (b) Relative error rate.

Figure 13a illustrates how the detection device's heading angular velocity changes with the track speed difference when traveling on a 5° slope at a center of mass linear velocity of 1 m/s. All three methods show a similar trend: the heading angular velocity increases with the increase in track speed difference.

Figure 13b presents the relative error rates between the virtual prototype simulation, physical prototype testing, and theoretical model calculations. The error rate increases with the track speed difference; however, the maximum relative error rate is 7.5%, which remains acceptable for current engineering standards. These results indicate that the proposed kinematic model demonstrates high accuracy and stability for turning on a slope.

Figure 14a shows the detection device's change in heading angular velocity as the track speed difference varies while traveling on a 10° slope at a center of mass linear velocity of 1.3 m/s.

Figure 14a indicates that the heading angular velocity trends are similar across the three methods, increasing with the track speed difference. Figure 14b shows the relative error

rates between the virtual prototype simulation, physical prototype testing, and kinematic model calculations. Although the physical prototype results exhibit some discrepancies, the relative error rate increases with the track speed difference. The maximum relative error rate is 7.5%, which is acceptable, meeting current engineering requirements. This demonstrates that the kinematic model has high accuracy and stability for turning on a slope.

5. Threshold Analysis of Stability Indicators for the Detection Device During Operation

5.1. Problem Analysis

When an electrically driven tracked intelligent detection device operates in unstructured and harsh outdoor environments, external factors such as uneven terrain and slippery surfaces can lead to instability and tipping during rapid turns. Common instability scenarios for the detection device are illustrated in Figure 15:

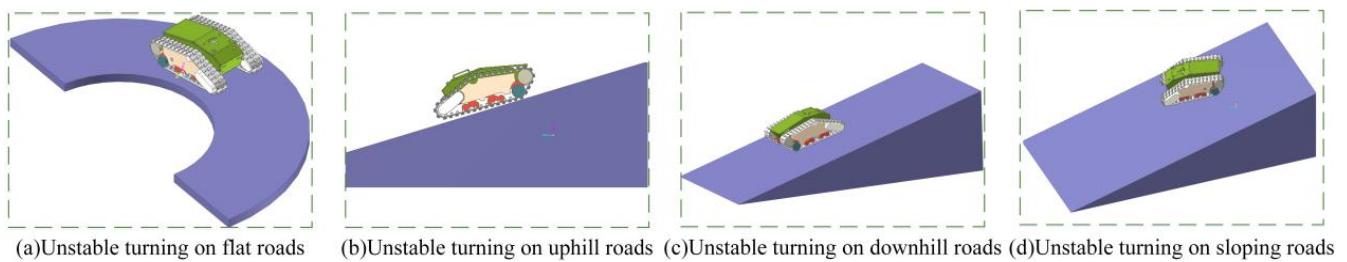


Figure 15. Classification of common instability conditions in tracked engineering vehicles. (a) Unstable turning on flat roads. (b) Unstable turning on uphill roads. (c) Unstable turning on downhill roads. (d) Unstable turning on sloping roads.

In Figure 15a, on flat terrain, excessive turning speed results in high lateral acceleration and yaw rate, increasing the risk of rollover. Figure 15b shows that during uphill movement, the rearward shift of the center of gravity reduces the front load, causing the front to lift, which can lead to instability due to “nose lifting”. Figure 15c depicts the “tail swing” phenomenon during downhill turns, where inertia can induce instability. In Figure 15d, under specific conditions, the coupling of steering and slope angles during rapid turns on a slope can cause large lateral accelerations, leading to instability [26].

5.2. Analysis Framework for Threshold Indicators of Driving Stability in Detection Devices Based on Multi-Model Driven Approaches

The classification method above only categorizes instability phenomena at a superficial level. Specific common characteristics precede such events regardless of the underlying causes of instability in detection devices. For instance, parameters such as lateral acceleration, yaw acceleration, roll rate, and roll angle can exceed specific thresholds, leading to instability. To better analyze detection devices’ motion characteristics and evaluation indicators, comprehensively considering the various factors affecting dynamic stability is essential. By integrating structural and operational characteristics, we can analyze the dynamic properties of detection devices under unstructured, adverse, and rapid steering conditions, thereby establishing an analytical framework as shown in Table 7.

5.3. Experimental Results and Discussion

To comprehensively analyze the detection device’s dynamic stability on a flat surface, we will evaluate its limit stability indicators at different angles, maintaining constant steering duration. Given the gap between the detection device and the simulated surface during the experiment, we will focus on its steady-state motion within a 1 to 10 s interval.

Table 7. A multi-model driven framework for analyzing the driving stability of electric-tracked intelligent detection devices.

A multi-model driven framework for analyzing the driving stability of electric-tracked intelligent detection devices
Develop a virtual prototype model; construct a dynamic model. Input: Drive motor model, ground parameters; Output: Dynamic characteristics parameters of the detection device; 1 Establish constraints for model assembly; 2 Crawler frame-drive wheel: revolute joint; crawler frame-drive wheel: revolute joint; crawler frame-tensioning device: prismatic joint; crawler frame-tensioning device: prismatic joint; tensioning device-guide wheel: revolute joint; 3 Optimize the model and adjust parameters; 4 Set ground parameters for the detection device to traverse a mixed surface of flat, gentle, and steep slopes. The surface is a hard pavement with a gradient of 15° and an adhesion coefficient of 0.85; 5 Establish simulation conditions; 6 Conduct simulations for various conditions; 7 Obtain dynamic characteristics of the detection device during an emergency maneuver; 8 Modify initial parameters and repeat the experiment; 9 Analyze dynamic characteristic data to determine stability metrics and their thresholds; 10 Construct a physical prototype of an electrically driven tracked intelligent detection device; 11. Configure high-speed turning conditions on the ground; 12 Integrate stability constraint metrics into the autonomous driving algorithm; 13 Verify the presence of instability; 14 Complete the driving stability analysis.

5.3.1. Stability Analysis of the Detection Device During Obstacle Avoidance on a Flat Surface

Figure 16 illustrates the stability analysis of the detection device during rapid obstacle avoidance maneuvers on a flat surface across various steering angles. From Figure 16a–c, it is evident that as the steering angle increases, the longitudinal, lateral, and vertical accelerations become increasingly pronounced. Notably, during a 60° steering maneuver, the peak instantaneous longitudinal acceleration a_x reaches 9.3 m/s², while the lateral acceleration a_y peaks at 28.4 m/s². Figure 16d–f indicate significant fluctuations in roll, pitch, and yaw angular accelerations with increasing steering angles. The maximum roll angular acceleration a_{rx} in this brief rapid steering interval is 95.4 rad/s², the maximum pitch angular acceleration a_{ry} is 27.4 rad/s², and the maximum yaw angular acceleration a_{rz} is 44.7 rad/s² (“+” and “−” denote direction only, not magnitude, as in subsequent analyses).

5.3.2. Stability Analysis of the Detection Device During Obstacle Avoidance on an Uphill Surface

Figure 17 presents the stability analysis of the detection device during rapid steering for obstacle avoidance on an uphill surface at various angles. From Figure 17a–c, the peak instantaneous longitudinal acceleration a_x reaches 7.7 m/s², lateral acceleration a_y reaches 42.1 m/s², and vertical acceleration a_z reaches −22 m/s². Figure 17d–f show that during this brief rapid steering, the maximum roll angular acceleration a_{rx} is −169 rad/s², the maximum pitch angular acceleration a_{ry} is 43.3 rad/s², and the maximum yaw angular acceleration a_{rz} is −40.8 rad/s².

5.3.3. Stability Analysis of the Detection Device During Obstacle Avoidance on a Downhill Curve

Figure 18 demonstrates the stability analysis of the detection device during rapid steering for obstacle avoidance on a 15° downhill surface at various angles. From Figure 18a–c, the peak instantaneous longitudinal acceleration a_x is 13.4 m/s², lateral acceleration a_y is 34.8 m/s², and vertical acceleration a_z is 12.1 m/s². Figure 18d–f indicate that during this brief rapid steering maneuver, the maximum roll angular acceleration a_{rx} is 120.1 rad/s², the maximum pitch angular acceleration a_{ry} is −16.9 rad/s², and the maximum yaw angular acceleration a_{rz} is 36.5 rad/s².

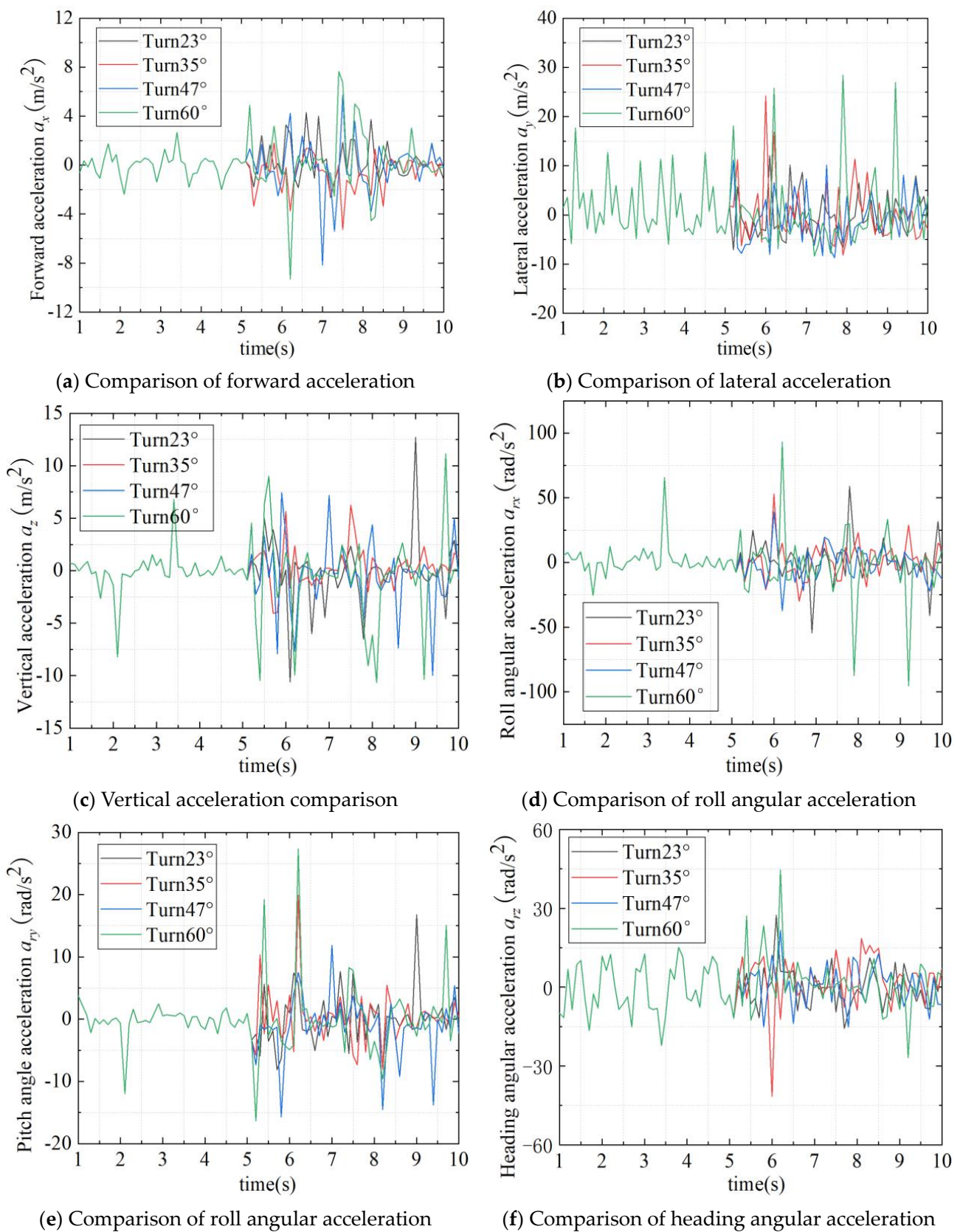
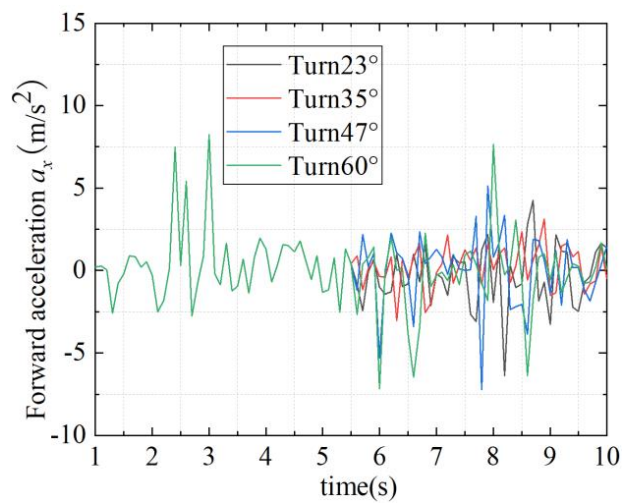
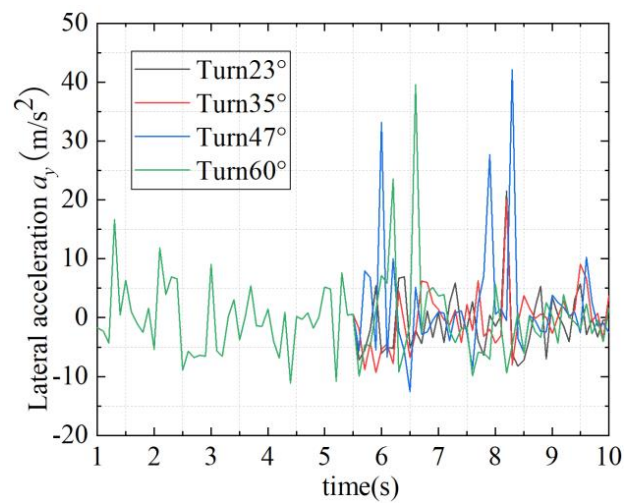


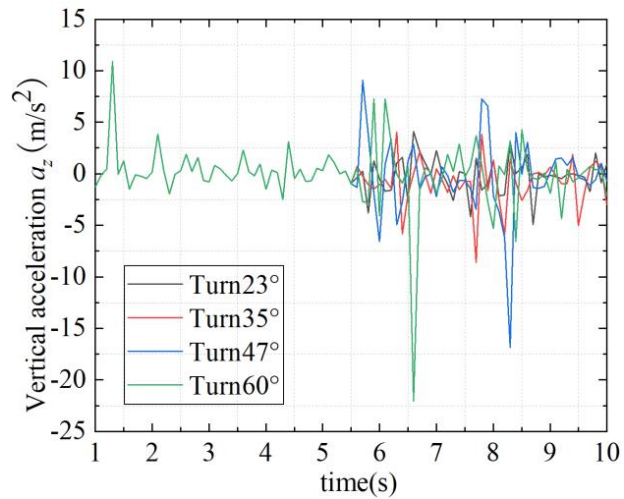
Figure 16. Flat road simulation analysis results.



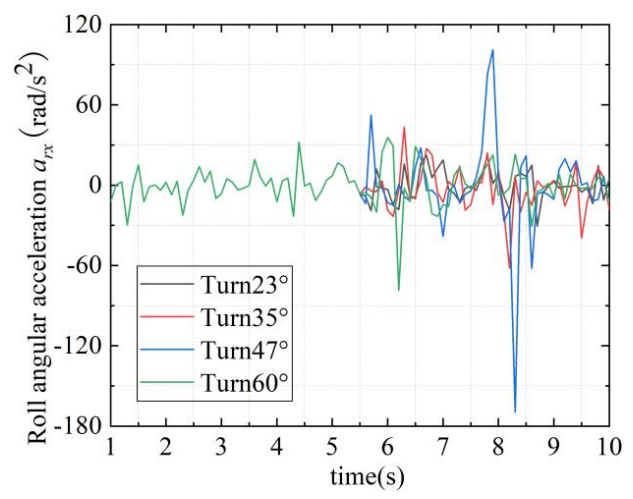
(a) Comparison of forward acceleration



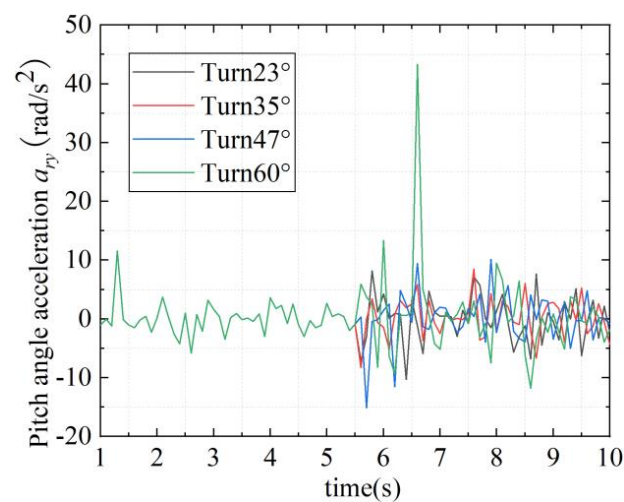
(b) Comparison of lateral acceleration



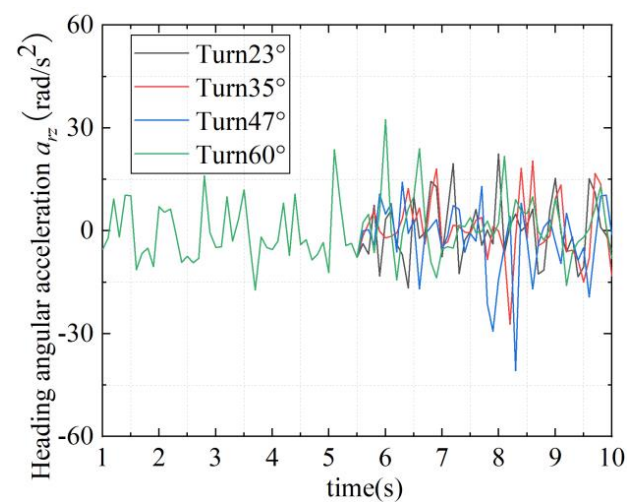
(c) Vertical acceleration comparison



(d) Comparison of roll angular acceleration



(e) Comparison of roll angular acceleration



(f) Comparison of heading angular acceleration

Figure 17. Uphill surface simulation analysis results.

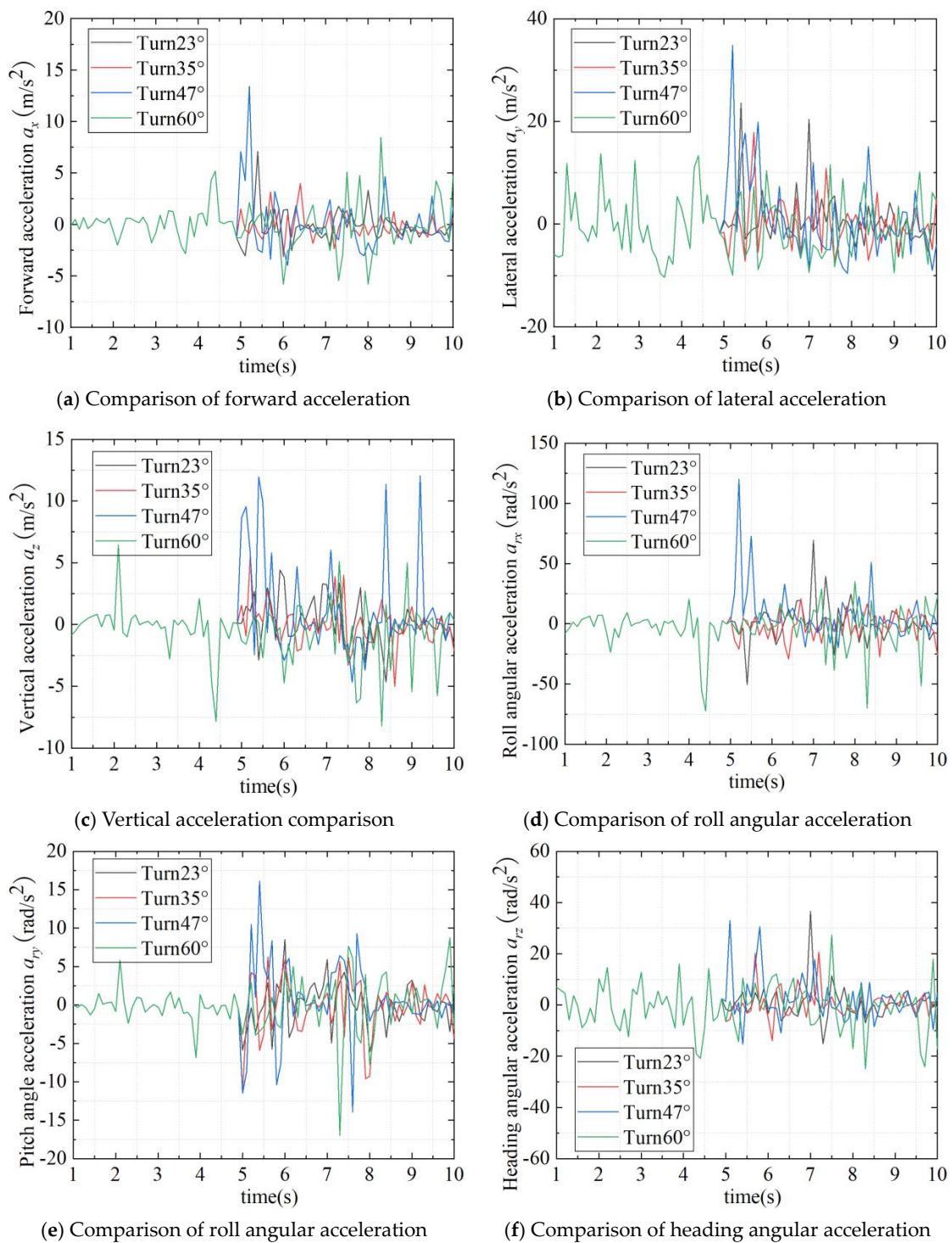
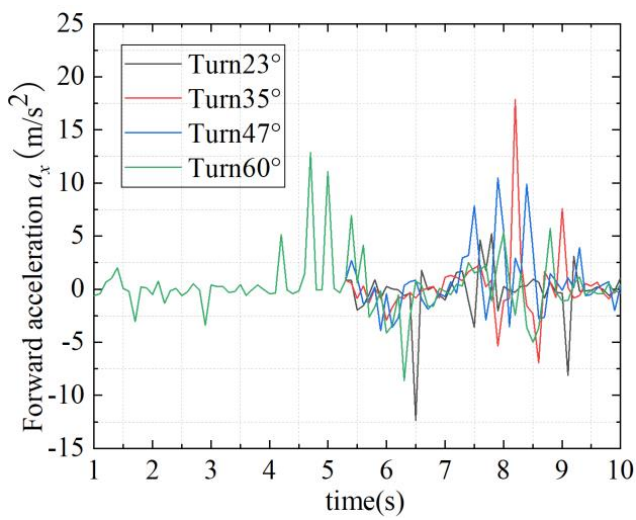


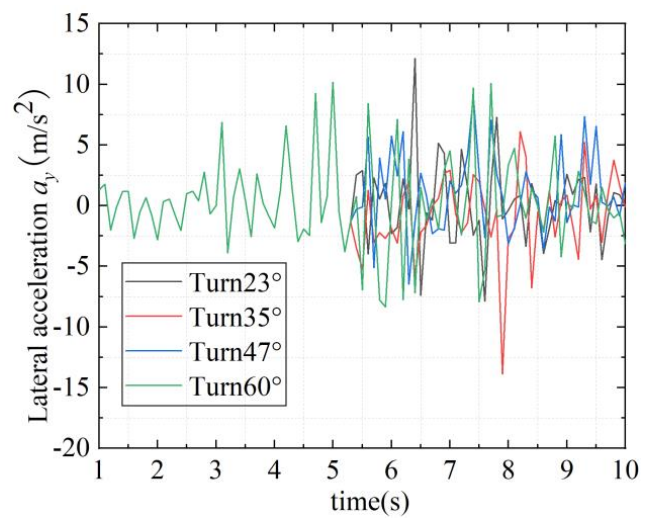
Figure 18. Downhill curve simulation analysis results.

5.3.4. Stability Analysis of the Detection Device During Obstacle Avoidance on a Sloped Curve

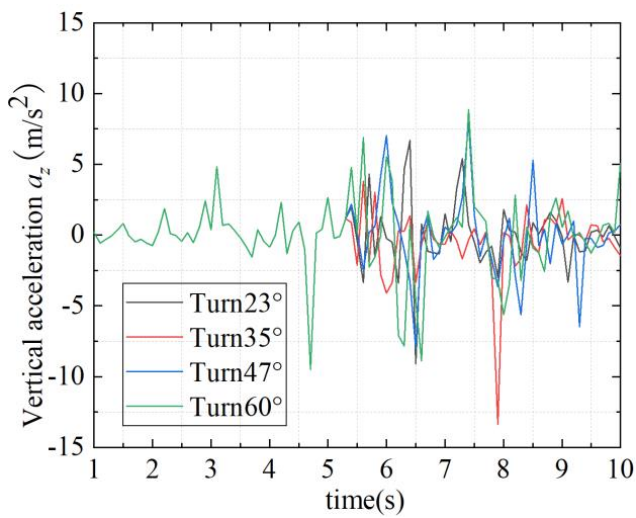
Figure 19 illustrates the stability analysis of the detection device during rapid steering for obstacle avoidance on a 15° gradient. From Figure 19a–c, the peak instantaneous longitudinal acceleration a_x is 17.9 m/s^2 , lateral acceleration a_y is −13.9 m/s^2 , and vertical acceleration a_z is −13.3 m/s^2 . Figure 19d–f show that during this brief rapid steering maneuver, the maximum roll angular acceleration a_{rx} is 53.3 rad/s^2 , the maximum pitch angular acceleration a_{ry} is −49.9 rad/s^2 , and the maximum yaw angular acceleration a_{rz} is −13.4 rad/s^2 .



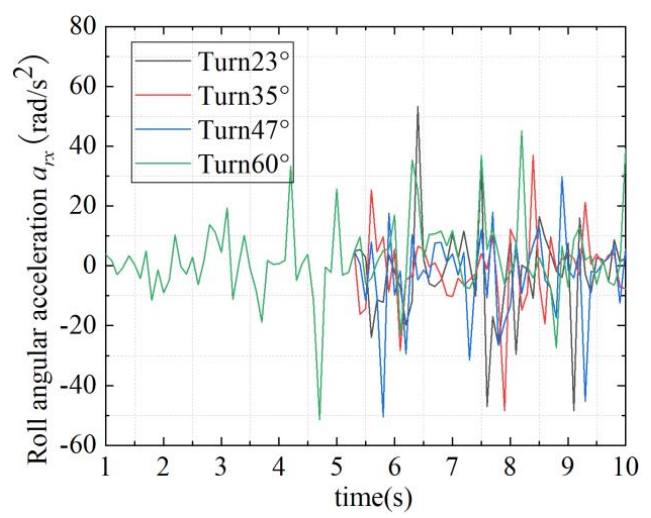
(a) Comparison of forward acceleration



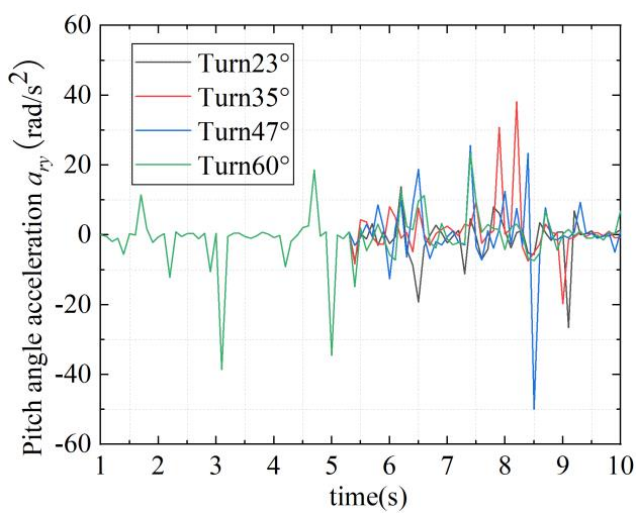
(b) Comparison of lateral acceleration



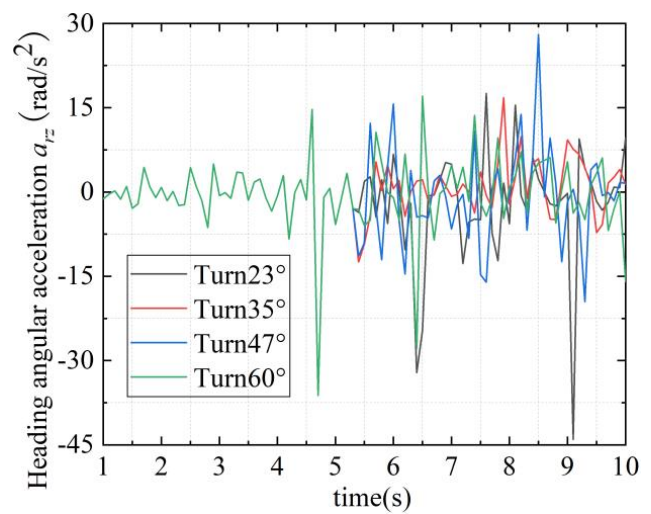
(c) Vertical acceleration comparison



(d) Comparison of roll angular acceleration



(e) Comparison of roll angular acceleration



(f) Comparison of heading angular acceleration

Figure 19. Sloped curve simulation analysis results.

Thus, the dynamic stability thresholds for the tracked intelligent mobile detection device are specified in Table 8.

Table 8. Dynamic stability thresholds for electrically driven tracked intelligent detection device.

Stability Indicators	a_x (g)	a_y (g)	a_z (g)	a_{rx} (rad/s ²)	a_{ry} (rad/s ²)	a_{rz} (rad/s ²)	DSE
Threshold	2	4.3	2.3	169	49.9	40.8	36.35

In summary, the dynamic stability index for the electrically driven tracked intelligent detection device can be represented based on its operational and slope information as follows:

$$\text{DSE} = \frac{\beta}{\beta_{\max}} \lambda \cdot \{a_x, a_y, a_z, a_{rx}, a_{ry}, a_{rz}\}^T \quad (31)$$

In Equation (31), a_x , a_y , and a_z represent the electrically driven tracked intelligent detection device's longitudinal, lateral, and vertical accelerations. a_{rx} , a_{ry} , and a_{rz} denote the pitch, roll, and yaw angular accelerations. β represents the angle of the slope where the detection device is located while driving on the slope; $\beta_{\max}$ is the maximum slope angle that the detection device can maintain during stable driving. This evaluation index integrates the motion state parameters of the detection device with the road slope angle. Additionally, $\lambda = \{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6\}$ is a weight vector, with the following relationship:

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 = 1 \quad (32)$$

When the detection device operates on unstructured roads, the lateral and longitudinal accelerations and roll and yaw angular accelerations are crucial for its driving stability. In contrast, vertical acceleration and pitch angular speed have a lesser impact. The results indicate that longitudinal and roll angular acceleration influence stability approximately twice as much as the latter. Therefore, the weight vector λ in this study is set as $\{0.2, 0.2, 0.1, 0.1, 0.2, 0.2\}$.

5.4. Ground Experiment

To comprehensively validate the effectiveness of the dynamic stability index in constraint control, we employed a physical prototype of an electrically driven tracked intelligent detection device and conducted ground tests under rapid steering conditions. The experimental results, depicted in the Figure 20, clearly show changes in the dynamic stability index. The data indicate that when the detection device employs constraint control based on the driving stability index, the maximum dynamic stability envelope (DSE) is effectively limited to a range of 15.8. This performance demonstrates that the device maintains excellent driving stability under extremely rapid steering conditions, thereby enormously validating the effectiveness and practicality of the applied constraint control strategy.

In the subsequent research stage, we hope to conduct theoretical calculations and analysis on the driving stability of detection devices in complex unstructured environments. Considering the influence of rainy and snowy weather, we aim to enhance the universality of the detection device in the real world by constructing a simulation dataset based on environmental slope, surface friction coefficient, and other environmental factors. This will help us to comprehensively study the stability analysis of construction vehicles under various road conditions.

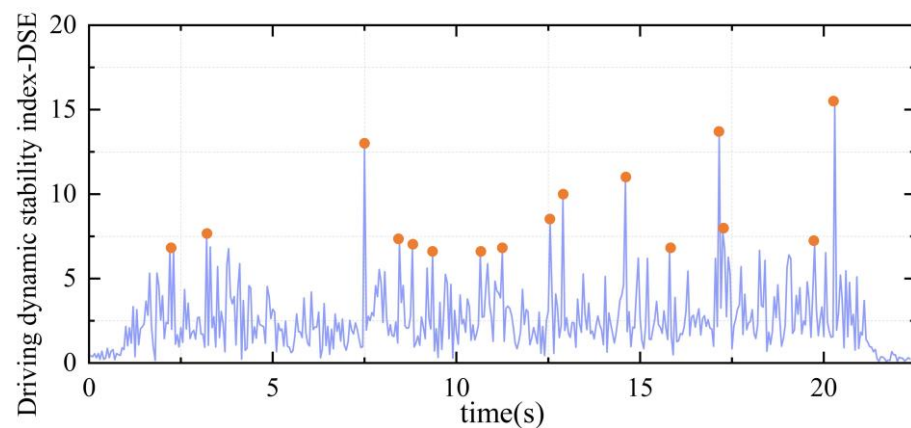


Figure 20. Ground experiment results.

6. Conclusions

This paper explores the kinematics and dynamics of tracked walking machinery by constructing kinematic and dynamic equations for an electrically driven tracked intelligent detection device. A virtual prototype and a physical prototype of the device were successfully developed. By combining theoretical analysis, virtual prototype simulation, and physical prototype testing, we systematically analyzed the stability of the tracked walking device as a carrier assembly under nonlinear multibody dynamic models, kinematic numerical models, virtual prototype models, and instability mechanisms in unstructured environments.

The main contributions of this study include:

1. Establishing dynamic equations for straight and steering conditions based on the structural and dynamic characteristics of the electrically driven tracked intelligent detection device. Kinematic models for straight and steering scenarios were constructed according to the device's driving modes. Through numerical simulation, virtual prototype simulation, and physical prototype testing, we analyzed the variation patterns of motion parameters such as centroid linear velocity, acceleration, steering radius, and heading angular velocity under typical driving conditions, thereby verifying the accuracy of the numerical model.
2. Enhancing dynamic stability indicators suitable for the device in common instability conditions in unstructured environments. Using multibody dynamics simulation software, RecurDyn (V9R4) and control software Matlab/Simulink (2014b), a virtual prototype simulation model was established to determine stability thresholds under common instability conditions through simulation.
3. Constructing a physical prototype and an on-site testing system for the electrically driven tracked intelligent detection device. Ground testing and analysis of the physical prototype validated the accuracy and feasibility of the proposed analytical methods.

In summary, utilizing kinematic and dynamic equations and virtual prototype analysis allows for a comprehensive assessment of dynamic characteristics and instability thresholds under various conditions, enhancing the device's walking stability and operational efficiency. This research provides valuable references for developing and designing electrically driven tracked intelligent detection devices and other engineering machinery.

In the next step of research, we will use high-precision sensors such as LiDAR to improve data acquisition accuracy. This, combined with the detection device's driving stability threshold, will enhance the detection device's stability and accuracy in unstructured environments, providing further solutions for the problem of autonomous driving in construction vehicles.

Author Contributions: Conceptualization, Y.W. and L.L.; methodology, Y.W.; software, Y.W.; validation, H.L. and C.L.; formal analysis, Y.Z.; investigation, C.L.; resources, L.L.; data curation, Y.W. and L.S.; writing—original draft preparation, S.L.; writing—review and editing, Z.Y.; visualization, L.S.;

supervision, Y.H.; project administration, S.L.; funding acquisition, L.L., X.Z. and M.X. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Jilin Province Science and Technology Development Plan, Project (No.20240302048GX). This research was funded by the Jilin Province and Changchun City Major Science and Technology Special, Project (No.20230301005ZD). This research was funded by the Scientific Research Project of the Education Department of Guangdong Province, Project (No.2022KCXTD029). This research was supported by the Jilin Province Postdoctoral Merit Funding (No.634230337003).

Data Availability Statement: The original contributions presented in the study are included in the article; further inquiries can be directed to the corresponding author.

Conflicts of Interest: Mr. Chunbai Liu and Mr. Yanzhong Hao were employed by FAW Casting Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

- Alexa, O.; Ciobotaru, T.; Grigore, L.S.; Grigorie, T.L.; Stefan, A.; Oncioiu, I.; Priescu, I.; Vladescu, C. A Review of Mathematical Models Used to Estimate Wheeled and Tracked Unmanned Ground Vehicle Kinematics and Dynamics. *Mathematics* **2023**, *11*, 3735. [\[CrossRef\]](#)
- Moreschi, C.; Da Broi, U.; Cividino, S.R.S.; Gubiani, R.; Pergher, G.; Vello, M.; Rinaldi, F. The analysis of the cause-effect relation between tractor overturns and traumatic lesions suffered by drivers and passengers: A crucial step in the reconstruction of accident dynamics and the improvement of prevention. *Agriculture* **2017**, *7*, 97. [\[CrossRef\]](#)
- Artuñedo, A.; Moreno-Gonzalez, M.; Villagra, J. Lateral control for autonomous vehicles: A comparative evaluation. *Annu. Rev. Control.* **2024**, *57*, 100910. [\[CrossRef\]](#)
- Zhu, Z.; Tang, X.; Qin, Y.; Luan, Q.; Luo, X.; Xu, G.; Liu, H.; Zhang, Y.; Wang, J.; Zhang, Q.; et al. A survey of lateral stability criterion and control application for autonomous vehicles. *IEEE Trans. Intell. Transp. Syst.* **2023**, *24*, 10382–10399. [\[CrossRef\]](#)
- Lei, T.; Wang, J.; Yao, Z. Modelling and stability analysis of articulated vehicles. *Appl. Sci.* **2021**, *11*, 3663. [\[CrossRef\]](#)
- Lai, F.; Huang, C.; Ye, X. Analysis of vehicle driving stability based on longitudinal-lateral and vertical unified dynamics model. *Int. J. Automot. Technol.* **2022**, *23*, 73–87. [\[CrossRef\]](#)
- Zhou, Y.Z.; Wang, R.C.; Ding, R.K.; Shi, D.H.; Ye, Q. Investigation on hierarchical control for driving stability and safety of intelligent HEV during car-following and lane-change process. *Sci. China Technol. Sci.* **2022**, *65*, 53–76. [\[CrossRef\]](#)
- Zhu, J.; Zhang, S.; Wang, G.; Zhang, W.; Zhang, S. Research on vehicle stability region under critical driving situations with static bifurcation theory. *Proceedings of the Institution of Mechanical Engineers, Part D. J. Automob. Eng.* **2021**, *235*, 2072–2085. [\[CrossRef\]](#)
- Liu, R.; Wie, M.; Sang, N.; Wie, J. A trajectory-tracking controller for improving the safety and stability of four-wheel steering autonomous vehicles. *Transport* **2021**, *36*, 147–163. [\[CrossRef\]](#)
- Yu, S.; Li, W.; Li, Y.; Chen, H.; Chu, H.; Lin, B.; Yu, J. Nonlinear Predictive Control of Active Four-wheel Steering Vehicles. *Int. J. Control. Autom. Syst.* **2023**, *21*, 3336–3347. [\[CrossRef\]](#)
- Nguyen, H.D.; Kim, D.; Son, Y.S.; Han, K. Linear time-varying mpc-based autonomous emergency steering control for collision avoidance. *IEEE Trans. Veh. Technol.* **2023**, *72*, 12713–12727. [\[CrossRef\]](#)
- Zhou, C.; Yu, L.; Li, Y.; Lu, Z.; Song, J. A layered roll stability control strategy for commercial vehicles based on adaptive model predictive control. *Veh. Syst. Dyn.* **2023**, *61*, 3067–3088. [\[CrossRef\]](#)
- Kati, M.S.; Fredriksson, J.; Jacobson, B.; Laine, L. A feedback-feed-forward steering control strategy for improving lateral dynamics stability of an A-double vehicle at high speeds. *Veh. Syst. Dyn.* **2022**, *60*, 3955–3976. [\[CrossRef\]](#)
- Guo, L.; Xu, H.; Zou, J.; Jie, H.; Zheng, G. A state observation and torque compensation-based acceleration slip regulation control approach for a four-wheel independent drive electric vehicle under slope driving. *Proceedings of the Institution of Mechanical Engineers, Part D. J. Automob. Eng.* **2020**, *234*, 2728–2743. [\[CrossRef\]](#)
- Haque, T.S.; Rahman, M.H.; Islam, M.R.; Razzak, A.; Badal, F.R.; Ahamed, H. A review on driving control issues for smart electric vehicles. *IEEE Access* **2021**, *9*, 135440–135472. [\[CrossRef\]](#)
- Kenda, J.; Kopac, J. Measurements and analyses of lateral acceleration in traffic of vehicles. *Teh. Vjesn.* **2011**, *18*, 281–286.
- Dandiwalla, A.; Chakraborty, B.; Chakravarty, D.; Sindha, J. Vehicle dynamics and active rollover stability control of an electric narrow three-wheeled vehicle: A review and concern towards improvement. *Veh. Syst. Dyn.* **2023**, *61*, 399–422. [\[CrossRef\]](#)
- Steinginga, L.; Huang, W.; Poole, G. Analytical and experimental investigation of roll stability of a truck towing a special purpose trailer with no suspension. *Int. J. Veh. Perform.* **2022**, *8*, 296–315. [\[CrossRef\]](#)
- Fu, W.; Wang, X.; Zhang, X. Rollover Stability of Heavy-Duty AGVs in Turns Considering Variation in Friction Coefficient. *Lubricants* **2023**, *11*, 119. [\[CrossRef\]](#)
- Wang, C.; Wang, Z.; Zhang, L.; Cao, D.; Dorrell, D.G. A vehicle rollover evaluation system based on enabling state and parameter estimation. *IEEE Trans. Ind. Inform.* **2020**, *17*, 4003–4013. [\[CrossRef\]](#)

21. Sukumaran, R.; Gaurkar, P.V.; Subramanian, S.C. Integrated Rollover Prevention and Antilock Brake System for Heavy Commercial Road Vehicles. *IEEE Access* **2023**, *11*, 124081–124097. [[CrossRef](#)]
22. Kim, Y.; Kwon, S.J. Robust stabilization of underactuated two-wheeled balancing vehicles on uncertain terrains with nonlinear-model-based disturbance compensation. *Actuators* **2022**, *11*, 339. [[CrossRef](#)]
23. Lee, J.; Jang, Y.H.; Kim, H.S.; Kim, S.-L.; Ko, S.-W. Enabling Distributed Control of Vehicle Platooning via Over-the-Air Consensus. *IEEE Trans. Wirel. Commun.* **2024**, *23*, 17205–17221. [[CrossRef](#)]
24. Lee, S.; Hwang, S.; Kim, H.S.T.-S. Fuzzy Observer-based Output Feedback Lateral Control of UGVs Using a Disturbance Observer. *Drones* **2024**, *8*, 685. [[CrossRef](#)]
25. Luo, J.; Wu, K.; Wang, Y.; Wang, T.; Zhang, G.; Liu, Y. An Improved UKF for IMU State Estimation Based on Modulation LSTM Neural Network. *IEEE Trans. Intell. Transp. Syst.* **2024**, *25*, 10702–10711. [[CrossRef](#)]
26. Wang, Y.; Lin, J.; Zhang, L.; Wang, T.; Xu, H.; Qi, Y. Stable Obstacle Avoidance Strategy for Crawler-Type Intelligent Transportation Vehicle in Non-Structural Environment Based on Attention-Learning. *IEEE Trans. Intell. Transp. Syst.* **2023**, *24*, 7813–7830. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.